

Problem Solving Tutorial 4 2025

Notes for tutors and model answers

To get the most out of the session, please attempt the questions marked **[Attempt before the PST]** in advance of the PST session. The tutor will work through the answers to these questions in the PST. The remaining questions (a past exam question) will be solved by your PST group in the class under the guidance of the tutor.

First and Second-order perturbation theory

Perturbation theory is an extremely powerful tool in theoretical physics, allowing us to extend our calculations (approximately) and also our intuitions from exactly solvable to non-exactly solvable systems. It is a key technique in many areas of physics, most notably atomic and molecular and particle physics.

The **aim** of this PST is for you to gain practice with the methods of non-degenerate perturbation theory (you will learn about degenerate perturbation theory in week 11's lecture - we will not cover it here).

1. **[Attempt before the PST]**. A spin half-particle sits a magnetic field in the Z direction, and has a Hamiltonian

$$\hat{H} = \hat{H}_0 = A\sigma_z$$

where A has units of energy. An additional magnetic field is applied at 45 degrees to the Z -axis, and the Hamiltonian changes to

$$\hat{H} = \hat{H}_0 + \hat{W} = A\sigma_z + \alpha(\sigma_x + \sigma_z)$$

- (a) Before you do any calculations, for what range of values of A and α would you expect that first order perturbation theory could give an accurate calculation?

Solution: Loosely speaking α/A needs to be small.

Precisely speaking, first order perturbation theory will be valid when the amplitude of the matrix elements of the perturbation is smaller than the difference between energies of the unperturbed Hamiltonian.

One further scenario where we could use perturbation theory, if $\alpha \gg A$ we could treat $\hat{H}_0$ as the Hamiltonian and $\hat{W}$ as the perturbation and use first order perturbation theory again!

- (b) Use perturbation theory to calculate the first order shifts to the energy eigenvalues.

Solution: First, we need to calculate the eigenvalues and eigenvectors of $\hat{H}_0$. To do this we use the matrix representation of $\hat{H}_0$.

$$A\sigma_z = A \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

We recall that the eigenvalues of a diagonal matrix are simply the diagonal entries, so the eigenvalues of H_0 are $E_1^0 = -A$ and $E_2^0 = +A$.

The corresponding eigenvectors are $|\psi_1^{(0)}\rangle$:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and $|\psi_2^{(0)}\rangle$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

respectively. We can either write down from our knowledge of the properties of diagonal matrices or could calculate directly.

The matrix representation of W is:

$$\alpha(\sigma_z + \sigma_x) = \alpha \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Hence the first order energy shifts are given by

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{W} | \psi_n^{(0)} \rangle$$

$$E_1^{(1)} = \alpha \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -\alpha$$

$$E_2^{(1)} = \alpha \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \alpha$$

and thus

$$E_1 \approx -A - \alpha$$

$$E_2 \approx A + \alpha$$

- (c) In this simple example, $\hat{H}$ is a 2x2 matrix, so we can compare the result of perturbation theory with the exact case. Calculate an exact expression for the eigenvalues of $\hat{H}$.

Solution: We need to find the eigenvalues of the matrix

$$A\sigma_z + \alpha(\sigma_x + \sigma_z) = A \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \alpha \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} A + \alpha & \alpha \\ \alpha & -(A + \alpha) \end{bmatrix}$$

Thus we solve

$$\left| \begin{bmatrix} A + \alpha - \lambda & \alpha \\ \alpha & -(A + \alpha) - \lambda \end{bmatrix} \right| = 0$$

From which, after a few simplifications we get:

$$\lambda = \pm \sqrt{(A + \alpha)^2 + \alpha^2}$$

(d) Now, assuming that $\alpha \ll A$ and using the binomial approximation:

$$(1 + x)^{1/2} \approx \left(1 + \frac{1}{2}x\right)$$

for $x \ll 1$, show that to first order in α , the perturbation theory energy shifts computed in part a match the result of your direct eigenvalue calculation. Which method was easier?

Solution:

$$\lambda = \pm(A + \alpha)\sqrt{1 + \frac{\alpha^2}{A + \alpha^2}}$$

$$\lambda = \pm(A + \alpha)\sqrt{1 + \frac{\alpha^2}{A + \alpha^2}}$$

$$\lambda \approx \pm(A + \alpha) \left(1 + \frac{1}{2} \frac{\alpha^2}{A + \alpha^2}\right)$$

If we truncate this to first order in α we have

$$\lambda \approx \pm(A + \alpha)$$

which matches the result from first order perturbation theory. Using perturbation theory led us to the result much more easily!

2. [Past exam question 2023-24]:

A particle in a harmonic potential has Hamiltonian

$$\hat{H}_0 = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

The potential then is modified by a weak linear field, resulting in the Hamiltonian;

$$\hat{H} = \hat{H}_0 + \hat{W} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) + \lambda \hat{x}.$$

- (a) Using perturbation theory, compute the first order corrections to the energy for all eigenstates of $\hat{H}_0$.

Solution:

We always begin a perturbation theory analysis by identifying the eigenvalues and eigenvectors of the solved or "unperturbed" part of the Hamiltonian.

In this case, we recognise that this is a quantum harmonic oscillator, and we recall that the eigenvectors and eigenvalues are $|n\rangle$ and $((n+1)/2)\hbar\omega$ respectively.

Using the notation of perturbation theory, the 0th order states and energies are:

$$|\psi_n^{(0)}\rangle = |n\rangle$$

$$E_n^{(0)} = \frac{n+1}{2}\hbar\omega$$

Now we can apply the formula for the first order correction (in the exam this formula will be available in the Materials booklet).

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{W} | \psi_n^{(0)} \rangle$$

where here, $\hat{W} = \lambda \hat{x}$. Note we are choosing to keep the λ scalar as part of the $\hat{W}$ operator. As stated in the lectures, it is correct both to do this and also to treat λ as a "parameter of smallness" outside of $\hat{W}$, as long as you remember to add the correct power of λ to the final energy shift (more on this below).

To calculate the correction, we need to express $\hat{x}$ in terms of raising and lowering operators:

The notes (and the Materials Booklet) don't give us this directly, instead they defined $\hat{a}_+$ and $\hat{a}_-$ in terms of $\hat{x}$ and $\hat{p}$:

$$\hat{a}_+ = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} - \frac{i}{\hbar \alpha} \hat{p} \right] \quad \text{and} \quad \hat{a}_- = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} + \frac{i}{\hbar \alpha} \hat{p} \right], \quad \text{where } \alpha = (m\omega/\hbar)^{1/2}$$

Adding the two expressions together we get:

$$\hat{a}_+ + \hat{a}_- = \frac{2}{\sqrt{2}} \alpha \hat{x}$$

Thus:

$$\hat{x} = \frac{\hat{a}_+ + \hat{a}_-}{\sqrt{2}\alpha}$$

and thus

$$\hat{W} = \frac{\lambda}{\sqrt{2\alpha}}(\hat{a}_+ + \hat{a}_-)$$

We can now calculate the first order energy correction:

$$\begin{aligned} E_n^{(1)} &= \langle \psi_n^{(0)} | \hat{W} | \psi_n^{(0)} \rangle \\ &= \langle n | \hat{W} | n \rangle \\ &= \frac{\lambda}{\sqrt{2\alpha}} \langle n | (\hat{a}_+ + \hat{a}_-) | n \rangle \\ &= \frac{\lambda}{\sqrt{2\alpha}} (\langle n | \hat{a}_+ | n \rangle + \langle n | \hat{a}_- | n \rangle) \\ &= \frac{\lambda}{\sqrt{2\alpha}} (\sqrt{n+1} \langle n | n+1 \rangle + \sqrt{n} \langle n | n-1 \rangle) \\ &= 0 \end{aligned}$$

where due to the orthogonality of the states $|n\rangle$, $|n+1\rangle$ and $|n-1\rangle$, the first order correction to the energy is zero.

When the first order correction is zero, this often means we need to calculate the second order corrections, which is done in the next question.

- (b) Compute the second order correction to the energy of *the ground state only*. (2nd order perturbation theory is covered in the video lecture on section 6.1.4 and in the notes).

Solution: Now we are asked to compute the second order correction for the ground state only. That restriction simplifies the calculation a little. Now our unperturbed state is $|0\rangle$, the harmonic oscillator ground state.

The expression for 2nd order corrections is complicated, and can be found in the notes, and in the exam in the Materials booklet.

It is:

$$E_n^{(2)} = \sum_{k \neq n} \frac{|W_{kn}|^2}{(E_n^{(0)} - E_k^{(0)})} = \sum_{k \neq n} \frac{|\langle \psi_k^{(0)} | \hat{W} | \psi_n^{(0)} \rangle|^2}{(E_n^{(0)} - E_k^{(0)})}.$$

We see that it comprises a sum over ratios between matrix elements of $\hat{W}$ (modulus squared): $|\langle \psi_k^{(0)} | \hat{W} | \psi_n^{(0)} \rangle|^2$, and unperturbed energy differences $E_n^{(0)} - E_k^{(0)}$.

For us $n = 0$, so the relevant squared matrix elements are:

$$|\langle k|\hat{W}|0\rangle|^2$$

where $k \neq 0$.

Let's compute these:

$$\begin{aligned} & |\langle k|\hat{W}|0\rangle|^2 \\ &= \left| \frac{\lambda}{\sqrt{2}\alpha} \langle k|(\hat{a}_+ + \hat{a}_-)|0\rangle \right|^2 \\ &= \frac{\lambda^2}{2\alpha^2} |\langle k|\hat{a}_+|0\rangle + \langle k|\hat{a}_-|0\rangle|^2 \\ &= \frac{\lambda^2}{2\alpha^2} |\langle k|1\rangle + 0|^2 \\ &= \frac{\lambda^2}{2\alpha^2} |\delta_{k,1}|^2 \\ &= \frac{\lambda^2}{2\alpha^2} \delta_{k,1} \end{aligned}$$

where we are using the fact that λ and α are real.

Substituting this into the sum simplifies it significantly:

$$\begin{aligned} & \frac{\lambda^2}{2\alpha^2} \sum_{k \neq 0} \frac{\delta_{k,1}}{(E_0^{(0)} - E_k^{(0)})} \\ &= \frac{\lambda^2}{2\alpha^2} \frac{1}{(E_0^{(0)} - E_1^{(0)})}. \end{aligned}$$

Now

$$E_0^{(0)} - E_1^{(0)} = \hbar\omega \left(\frac{1}{2} - \frac{3}{2} \right) = -\hbar\omega$$

i.e. energy difference between unperturbed states $|0\rangle$ and $|1\rangle$ is $\hbar\omega$, but note the minus sign, we are subtracting the larger energy from the smaller one.

Hence the second order energy shift for the ground state energy is:

$$E_0^{(2)} = -\frac{\lambda^2}{2\alpha^2 \hbar\omega}$$

If you like, you can substitute $\alpha^2 = (m\omega/\hbar)$, to get:

$$E_0^{(2)} = -\frac{\lambda^2}{2m\omega^2}$$